

1. (Fall 2024, #2) Find all values of  $k$  and  $a$  so that

$$f(x) = \begin{cases} \frac{3 \sin(kx^2)}{x}, & x > 0 \\ 6 - 3k + ax, & x \leq 0 \end{cases}$$

is continuous and differentiable at  $x = 0$ .

continuity: need  $\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x) = f(0)$

①                      ②                      ③

①:  $\lim_{x \rightarrow 0^-} (6 - 3k + ax) = 6 - 3k$

②:  $\lim_{x \rightarrow 0^+} \frac{3 \sin(kx^2)}{x} \cdot (kx^2) = \lim_{x \rightarrow 0^+} 3kx = 0$

③:  $f(0) = 6 - 3k$

So  $① = ② = ③ \Rightarrow \boxed{k = 2}$

Now  $f(x) = \begin{cases} \frac{3 \sin(2x^2)}{x} & x > 0 \\ ax & x \leq 0 \end{cases}$

differentiability: need  $\lim_{x \rightarrow 0^-} \frac{f(x) - f(0)}{x} = \lim_{x \rightarrow 0^+} \frac{f(x) - f(0)}{x}$

④                      ⑤

④:  $\lim_{x \rightarrow 0^-} \frac{ax - 0}{x} = a$

⑤:  $\lim_{x \rightarrow 0^+} \frac{\frac{3 \sin(2x^2)}{x} - 0}{x} = \lim_{x \rightarrow 0^+} \frac{3 \sin(2x^2)}{2 \cdot x^2} \cdot 2 = 6$

④ = ⑤  $\Rightarrow \boxed{a = 6}$

2. (Fall 2023, #2) Consider the piecewise function

$$f(x) = \begin{cases} x^2 \sin \frac{1}{x^2} + 3 & \text{for } x \neq 0 \\ 3 & \text{for } x = 0 \end{cases}$$

(a) Is  $f$  continuous at  $x = 0$ ? Justify.

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \left( x^2 \sin \frac{1}{x^2} + 3 \right)$$

$$-1 \leq \sin \frac{1}{x^2} \leq 1 \quad \text{for all } x \neq 0$$

$$-x^2 \leq x^2 \sin \frac{1}{x^2} \leq x^2 \quad \text{for all } x \neq 0$$

$$-x^2 + 3 \leq x^2 \sin \frac{1}{x^2} + 3 \leq x^2 + 3 \quad \text{for all } x \neq 0$$

$$3 = \lim_{x \rightarrow 0} (-x^2 + 3) \leq \lim_{x \rightarrow 0} \left( x^2 \sin \frac{1}{x^2} + 3 \right) \leq \lim_{x \rightarrow 0} (x^2 + 3) = 3$$

So  $\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \left( x^2 \sin \frac{1}{x^2} + 3 \right) = 3$  by the squeeze theorem

Note  $f(0) = 3 = \lim_{x \rightarrow 0} f(x) \Rightarrow \boxed{f \text{ is continuous at } 3.}$

(b) Is  $f$  differentiable at  $x = 0$ ? (Justify). If it is, find the value of  $f'(0)$ .

$$\lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x} = \lim_{x \rightarrow 0} \frac{x^2 \sin \frac{1}{x^2} + 3 - 3}{x} = \lim_{x \rightarrow 0} x \sin \frac{1}{x^2}$$

$$-1 \leq \sin \frac{1}{x^2} \leq 1 \quad \text{for all } x \neq 0$$

$$-|x| \leq x \sin \frac{1}{x^2} \leq |x| \quad \text{for all } x \neq 0$$

$$0 = \lim_{x \rightarrow 0} -|x| \leq \lim_{x \rightarrow 0} x \sin \frac{1}{x^2} \leq \lim_{x \rightarrow 0} |x| = 0$$

So  $\lim_{x \rightarrow 0} x \sin \frac{1}{x^2} = 0$ . Since the limit exists,  $\boxed{f \text{ is differentiable at } x=0.}$

And  $\boxed{f'(0) = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x} = 0}$

3. (Spring 2023, #2) The gravitational force,  $F$ , exerted by a planet of (constant) radius  $R$  on an object a distance  $x$  away from the center of the planet is given by the function

$$F(x) = \begin{cases} \frac{kx}{R^3} & \text{if } x \leq R \\ \frac{k}{x^2} & \text{if } x > R \end{cases}$$

where  $k$  is some constant.

- (a) Use the definition of continuity to determine if  $F(x)$  is continuous at  $R$ .

$$\begin{aligned} \lim_{x \rightarrow R^+} F(x) &= \lim_{x \rightarrow R^+} \frac{k}{x^2} = \frac{k}{R^2} \\ \lim_{x \rightarrow R^-} F(x) &= \lim_{x \rightarrow R^-} \frac{kx}{R^3} = \frac{k}{R^2} \\ F(R) &= \frac{k}{R^2} \end{aligned} \quad \text{equal} \Rightarrow \boxed{F \text{ is continuous at } R}$$

- (b) Use the limit definition of the derivative to find an expression for  $F'(x)$  when  $x > R$ .

$$\begin{aligned} \text{For } x > R, \quad F(x) &= \frac{k}{x^2} \\ F'(x) &= \lim_{t \rightarrow x} \frac{F(t) - F(x)}{t - x} = \lim_{t \rightarrow x} \frac{\frac{k}{t^2} - \frac{k}{x^2}}{t - x} = \lim_{t \rightarrow x} \frac{\frac{k(x^2 - t^2)}{t^2 x^2}}{t - x} \\ &= \lim_{t \rightarrow x} \frac{k(x+t)(x-t)}{t^2 x^2 (t-x)} = \lim_{t \rightarrow x} \frac{-k(x+t) \cancel{(t-x)}}{t^2 x^2 \cancel{(t-x)}} = \frac{-k(2x)}{x^2 x^2} = \boxed{-\frac{2k}{x^3}} \end{aligned}$$

4. Spring 2019, #3 Find the derivative of the following function **using the definition**. No credit will be given for using derivative rules.

$$\begin{aligned} f'(x) &= \lim_{t \rightarrow x} \frac{f(t) - f(x)}{t - x} = \lim_{t \rightarrow x} \frac{\frac{1}{1+t^2} - \frac{1}{1+x^2}}{t - x} \\ &= \lim_{t \rightarrow x} \frac{\frac{(1+x^2) - (1+t^2)}{(1+t^2)(1+x^2)}}{t - x} \\ &= \lim_{t \rightarrow x} \frac{x^2 - t^2}{(t-x)(1+t^2)(1+x^2)} \\ &= \lim_{t \rightarrow x} \frac{(x+t)(x-t)}{(t-x)(1+t^2)(1+x^2)} \\ &= \lim_{t \rightarrow x} \frac{-(x+t)(\cancel{t-x})}{(\cancel{t-x})(1+t^2)(1+x^2)} \\ &= \lim_{t \rightarrow x} \frac{-(x+t)}{(1+t^2)(1+x^2)} \\ &= \boxed{\frac{-2x}{(1+x^2)^2}} \end{aligned}$$

5. (Fall 2018, #4) Let  $f(x) = \begin{cases} x^2 \cos\left(\frac{\pi}{x}\right), & x \neq 0 \\ 0, & x = 0 \end{cases}$

(a) Show that  $f(x)$  is continuous at  $x = 0$ .

$$-1 \leq \cos\left(\frac{\pi}{x}\right) \leq 1$$

$$-x^2 \leq x^2 \cos\left(\frac{\pi}{x}\right) \leq x^2$$

$$\lim_{x \rightarrow 0} \downarrow$$

$$\downarrow \lim_{x \rightarrow 0}$$

$$0 \leq \lim_{x \rightarrow 0} x^2 \cos\left(\frac{\pi}{x}\right) \leq 0$$

by Squeeze Theorem

So  $\lim_{x \rightarrow 0} f(x) = 0$  and  $f(0) = 0 \rightarrow f(x)$  is continuous at  $x = 0$ .

(b) Is  $f(x)$  differentiable at  $x = 0$ ?

$$f'(0) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{h^2 \cos\left(\frac{\pi}{h}\right) - 0}{h} = \lim_{h \rightarrow 0} h \cos\left(\frac{\pi}{h}\right)$$

$$-1 \leq \cos\left(\frac{\pi}{h}\right) \leq 1$$

$$-|h| \leq |h| \cos\left(\frac{\pi}{h}\right) \leq |h|$$

$$\lim_{h \rightarrow 0} \downarrow$$

$$\downarrow \lim_{h \rightarrow 0}$$

$$0 \leq \lim_{h \rightarrow 0} |h| \cos\left(\frac{\pi}{h}\right) \leq 0$$

So by Squeeze Theorem,  $\lim_{h \rightarrow 0} |h| \cos\left(\frac{\pi}{h}\right) = 0$

$$\rightarrow \lim_{h \rightarrow 0^+} h \cos\left(\frac{\pi}{h}\right) = 0 \quad \text{and} \quad \lim_{h \rightarrow 0^-} h \cos\left(\frac{\pi}{h}\right) = \lim_{h \rightarrow 0^-} -|h| \cos\left(\frac{\pi}{h}\right) = 0$$

So  $f'(0) = \lim_{h \rightarrow 0} h \cos\left(\frac{\pi}{h}\right) = 0 \rightarrow f(x)$  is differentiable at  $x = 0$ .